

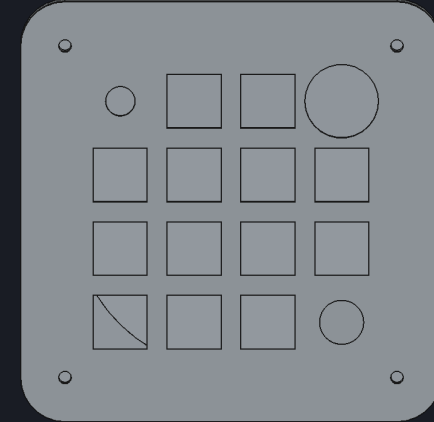
VibeKey Rev A

Two-device bill of materials

16 dynamically mapped inputs

Wireless + USB-C • 1,000 mAh • per-key RGB

Prototype procurement decision – 28 July 2026



What we are building

The BOM follows the current square enclosure and 16-input architecture

Controls

- 13 tactile MX keys
- 1 analog joystick + push
- 1 rotary encoder + push
- 1 capacitive touch region

Industrial design

- 114 × 114 mm square shell
- 6 mm corner radius
- 4° key-plane slope
- 86 mm circular bottom puck

User experience

- Every input remappable
- BLE and wired USB
- 13 individually addressed RGB keys
- Clear caps + one ceramic accent

The supplied visual sketch is an aesthetic reference; the electrical count remains the previously specified 4×4 set of sixteen logical inputs.

Electronics selection

Exact primary components for each of the two devices

Compute + radio

Raytac MDBT50Q-1MV2
nRF52840 certified module

Power path

TI BQ24074 charger
TPS63031 3.3 V buck-boost

RGB power

TPS61023 5 V boost
TPS22918 switch + AHCT buffer

Sensors

Alps RKJX21224001 joystick
AT42QT1010 touch
MAX17048 fuel gauge

Connectivity

GCT USB4105 USB-C
TI USB ESD + Nexperia VBUS TVS

Battery

TinyCircuits ASR00012
Protected 1,000 mAh; 1 A max

Exactly two devices

No third assembled unit and no discretionary spare quantities

26 each

Kailh Polia switches
Kailh hot-swap sockets
1N4148W matrix diodes
12 mA SK6812MINI-E LEDs

2 each

Radio modules • batteries
USB-C • joystick • encoder
Every power and sensor IC
JST-SH battery connectors

Pack minimums

2 × Adafruit 5068 12-cap packs
1 × Cerakey four-cap pack
1 silicone sheet + adhesive
5 bare PCBs may be fab minimum

JLC may require attrition quantities for placement. Those are manufacturing consumables—not parts for a third device.

The machine-readable CSV is the purchasing authority for quantities.

Who assembles what

Designed around no at-home reflow

JLCPCB / contract manufacturer

- All SMT passives and ICs
- Raytac radio module
- BQ24074 QFN charger
- Power converters and protection
- 13 reverse-mounted RGB LEDs
- 13 hot-swap sockets + diodes
- USB-C hybrid assembly preferred

Hand assembly

- Install encoder and joystick
- Insert MX switches and keycaps
- Connect protected battery
- Install heat-set inserts and screws
- Fit printed enclosure parts
- Cut/apply circular silicone pad
- Program and run acceptance test

Power budget and non-negotiable limits

RGB budget

13 × 12 mA channels
≈169 mA total at 5 V incl. static
≈313 mA battery draw at 3.0 V
Firmware cap: 156 mA channels

Battery + charging

1,000 mAh protected cell
1 A maximum discharge
Target total draw <0.45 A
Charge limited to ≈0.4 A

Thermal constraint

BQ24074 may dissipate ≈0.8 W
Thermal copper/vias required
Closed-case charge test mandatory
No USB-PD claim

Do not substitute generic 60 mA-per-pixel LEDs.

The RGB rail defaults off and is enabled only after controlled startup.

Two-device prototype budget

Planning allowance; final price comes from live JLC and distributor carts

PCBs + assembly

\$110-190

Five bare boards may be the fabrication minimum; p

Controls + mechanics

\$90-170

Switches, caps, joysticks, encoders, batteries, hardware

Shipping + contingency

\$50-100

Multiple vendors, taxes, rare-part loading, and first-article rev

\$250-460 total • \$125-230 per working prototype

Before placing the complete order

The BOM is selected; product validation still depends on the PCB and physical prototype

Complete schematic and four-layer routing; pass ERC/DRC and independent review.

Replace every placeholder with its exact manufacturer footprint and STEP model.

Redesign battery pocket, joystick opening/tail path, USB opening, and insert bosses.

Verify radio antenna keep-out against copper, battery, screws, and circular base.

Validate every JLC MPN, side, rotation, and package in the live order preview.

Build two units; test power, charging temperature, USB, BLE range, RGB, touch, all controls, and enclosure fit.

Recommendation

Buy now

Two batteries for fit checks
Two joysticks and encoders
Two keycap packs + ceramic pack
Switches and mechanical samples
Programming cable / nRF52840-DK if needed

Hold until layout validation

JLC assembled PCB order
Final printed enclosure batch
Custom-cut production foot pads
Any volume purchase beyond the two devices

Procurement source: hardware/bom.csv

Engineering rationale and gates: hardware/BOM.md